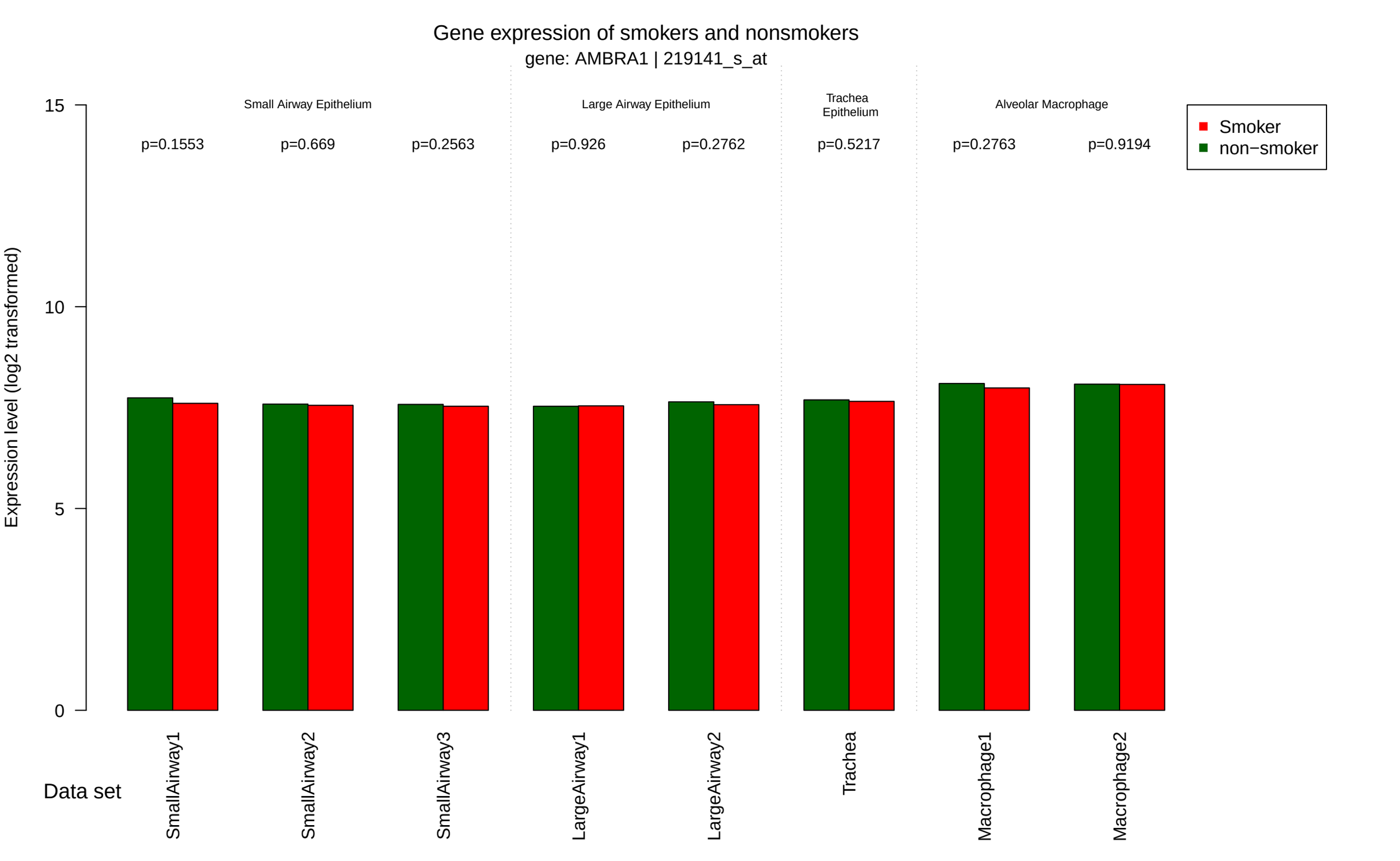
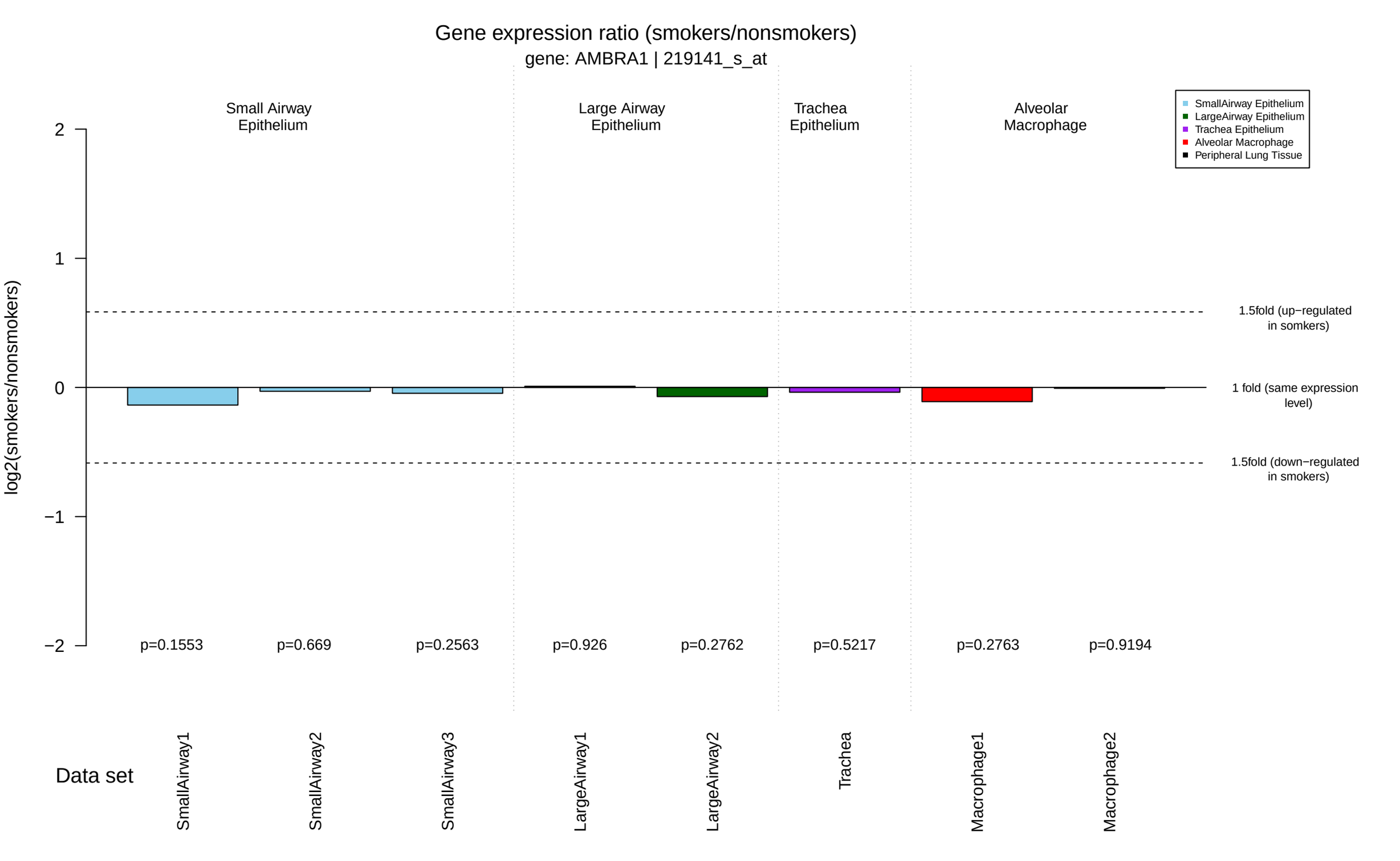
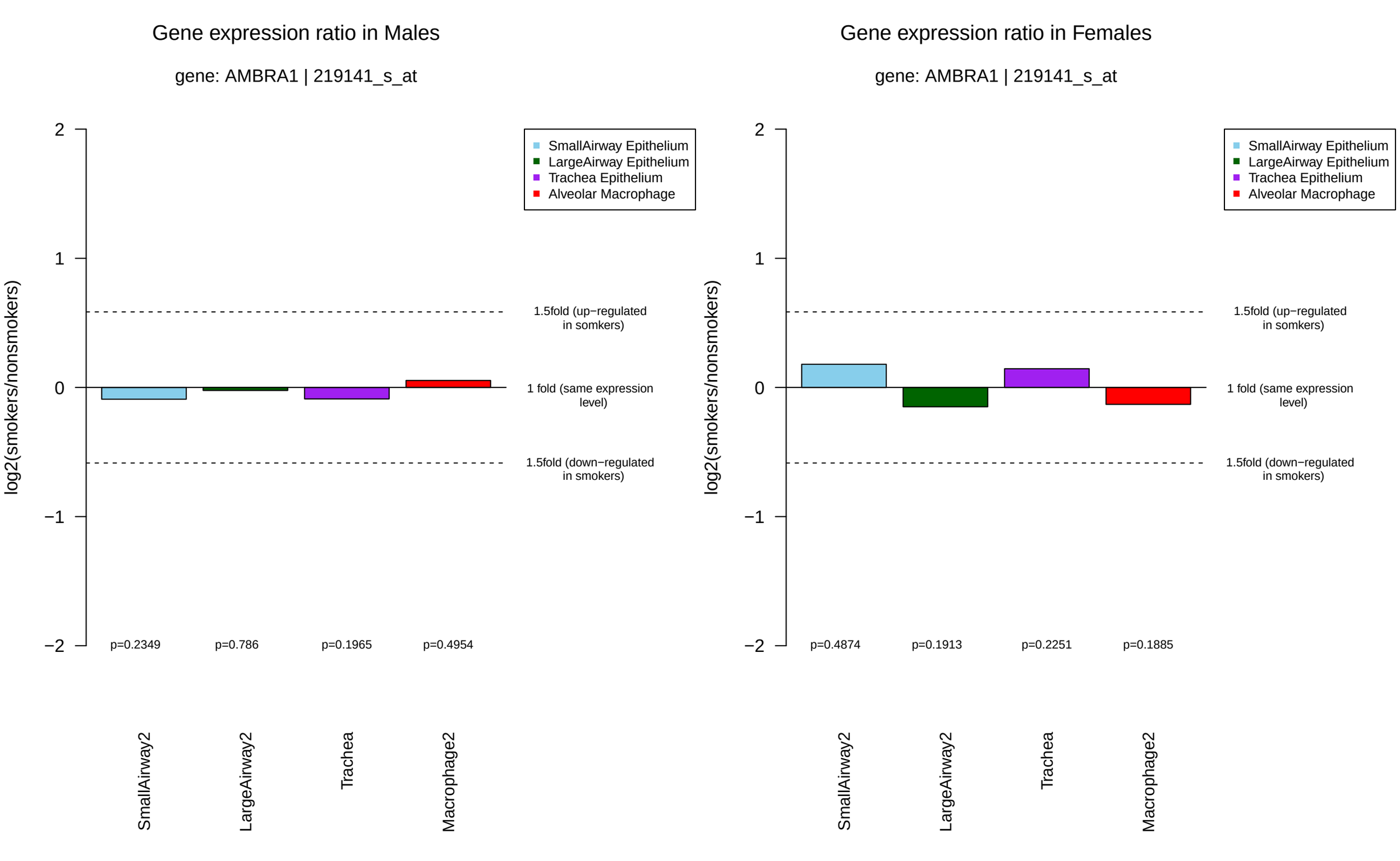
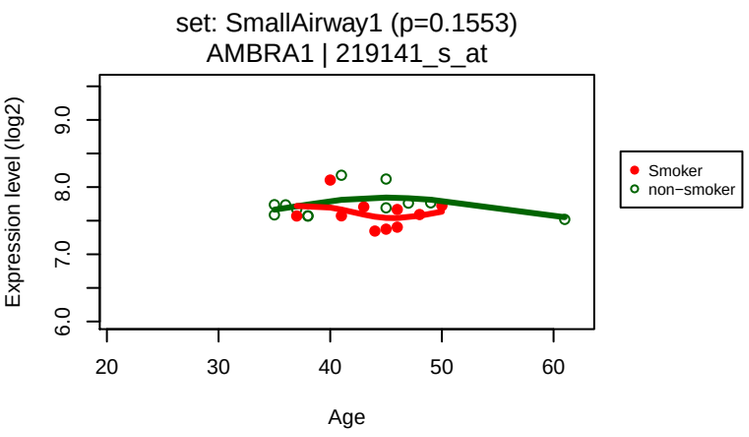
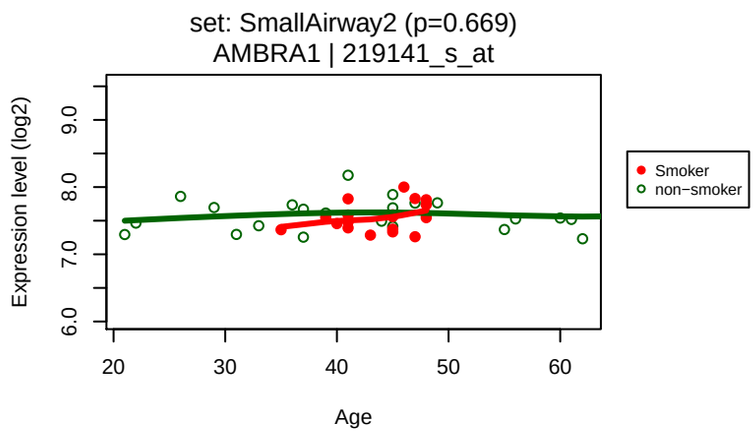
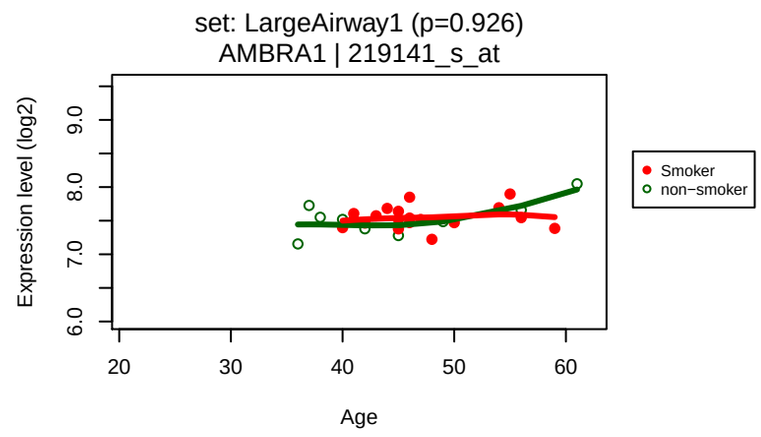

Gene expression of smokers and nonsmokers
gene: AMBRA1 | 219141_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1553
p=0.669
p=0.2563
p=0.926
p=0.2762
p=0.5217
p=0.2763
p=0.9194
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

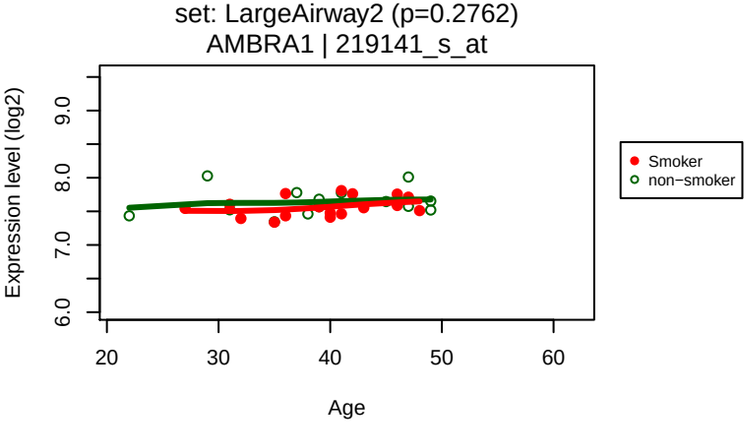

Gene expression ratio (smokers/nonsmokers)
gene: AMBRA1 | 219141_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1553
p=0.669
p=0.2563
p=0.926
p=0.2762
p=0.5217
p=0.2763
p=0.9194
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

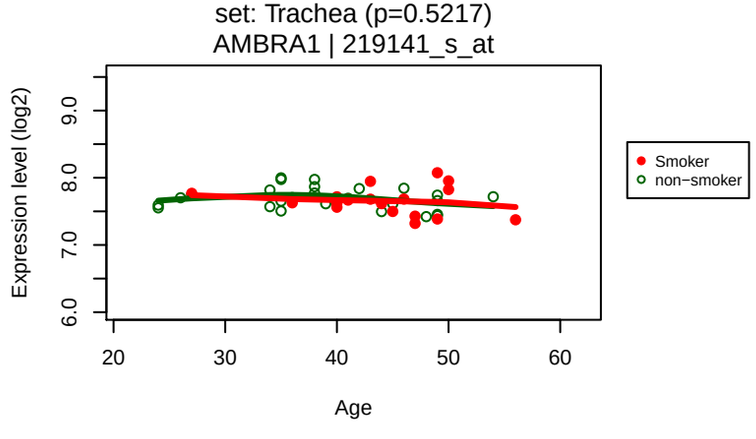

Gene expression ratio in Males
gene: AMBRA1 | 219141_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2349
p=0.786
p=0.1965
p=0.4954
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AMBRA1 | 219141_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4874
p=0.1913
p=0.2251
p=0.1885
SmallAirway2
LargeAirway2
Trachea
Macrophage2

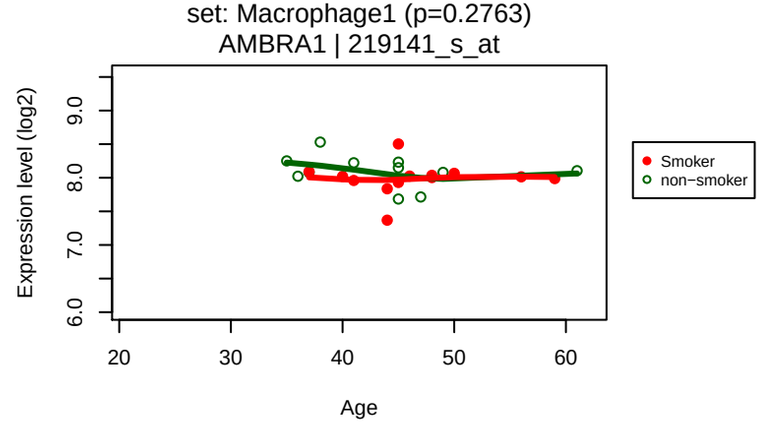

set: SmallAirway1 (p=0.1553)
AMBRA1 | 219141_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

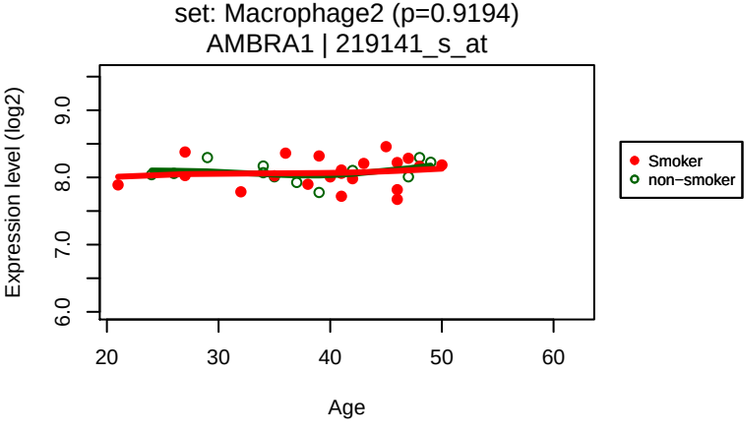

set: SmallAirway2 (p=0.669)
AMBRA1 | 219141_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

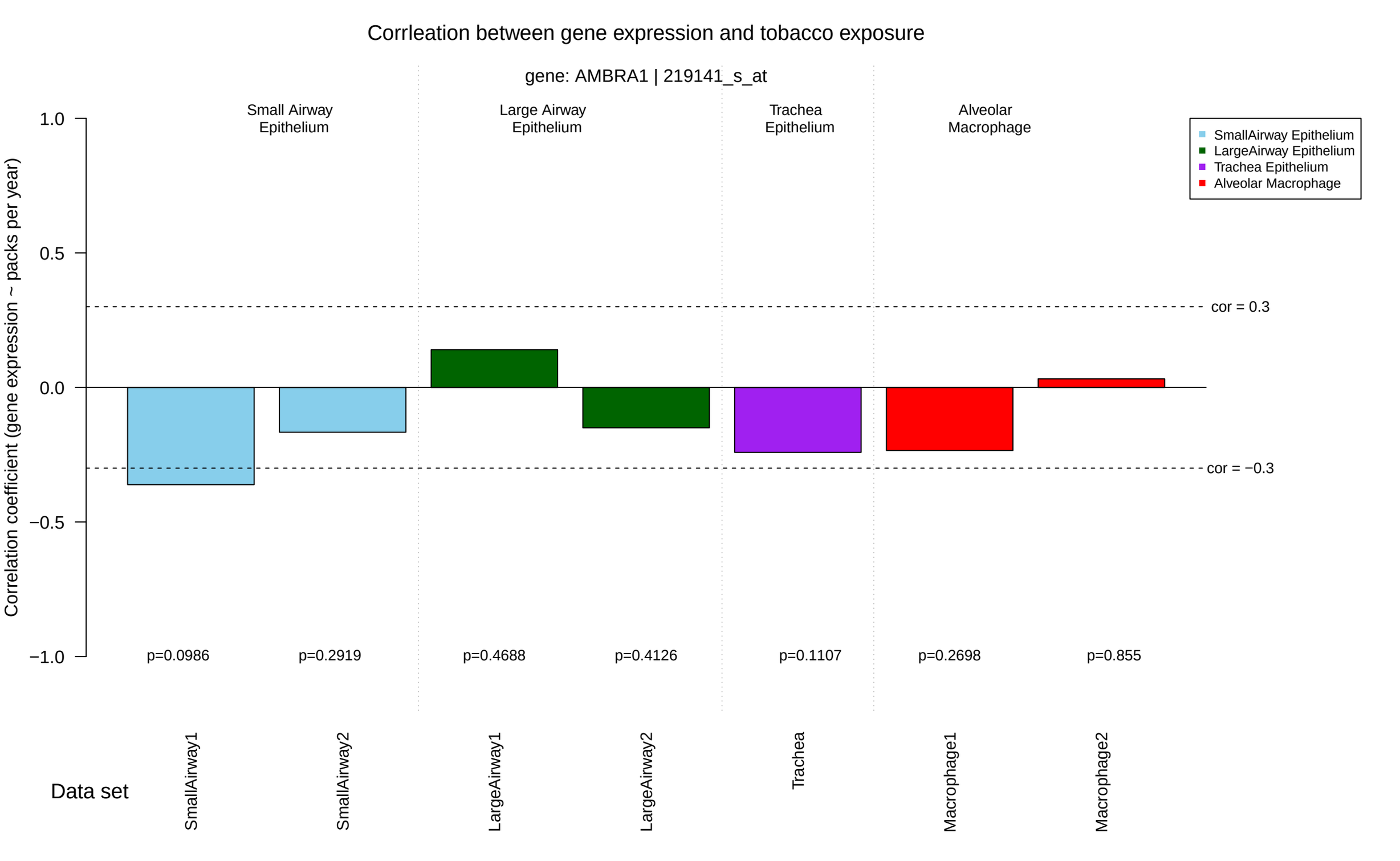

set: LargeAirway1 (p=0.926)
AMBRA1 | 219141_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2762)
AMBRA1 | 219141_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.5217)
AMBRA1 | 219141_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2763)
AMBRA1 | 219141_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9194)
AMBRA1 | 219141_s_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AMBRA1 | 219141_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0986
p=0.2919
p=0.4688
p=0.4126
p=0.1107
p=0.2698
p=0.855
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2